Olist E-Commerce Analytics

Team Name · Oky

TEAM MEMBERS

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Three problems eroding performance

Delivery delays, geographic concentration, and a satisfaction gap are compounding into structural risk. Visible across 96,469 orders on the Olist platform.

6.8%

Late delivery rate — ~6,560 customers per year receive orders past the promised date

~40%

Of all orders concentrated in São Paulo — a single-region dependency creating geographic risk

37%

Satisfaction gap between fast and slow deliveries — review scores of 4.6 vs 2.9 out of 5

Platform overview

BRL 15M
Total revenue, 96,469 orders

80% credit card
BRL 12.1M via card — instalment-friendly mix

~40% São Paulo
Share of all platform orders

12.1-day avg
Delivery vs. 3–7 day benchmark

4.2 / 5
Avg review — 0.3 below repeat-purchase threshold

6.8% late rate
Sep 2016 – Oct 2018 dataset

Dataset: Olist Brazilian E-Commerce, Sep 2016 – Oct 2018. Revenue in BRL.

01 ETL pipeline

Python (Pandas, NumPy): feature engineering, outlier treatment, and type standardisation across 96,469 real orders.

02 Statistical analysis

EDA and statistical modelling in Jupyter Notebooks, producing 10 decision-ready insights.

03 Tableau dashboard

Executive KPI view and operational drill-down with state, date, and category filters.
Open Tableau Dashboard →

04 ML models

Late delivery risk classifier (79.4% accuracy) and K-Means customer segmentation, both in production.
View on GitHub →

05 Live web app

React on Vercel and FastAPI on Render — fully accessible end-to-end for stakeholders.
Visit the app →

ML model results

What the ML models found

A binary classifier flags late-delivery risk at order placement. K-Means segments customers into four CRM-actionable profiles.

Delivery risk

Flags late delivery at order placement. High-risk states: MA 82%, RJ 79%, BA 73%. Full 12-day window to intervene.

At-risk

Single purchase, experienced a delay, left a low review. Immediate win-back CRM action required.

High-value loyal

High spend, strong review scores, low delay sensitivity. Priority retention and upsell target.

Two more segments

Low-engagement one-time buyers and instalment-heavy mid-value customers — each with a distinct reactivation strategy.

Five recommendations with estimated impact

ML late delivery

Deploy ML prediction to target a 30% reduction in late deliveries (~1,968 orders recovered/year). Lifts avg review score by 0.1–0.2 points.

Geographic expansion

Focus on MG, RJ, and RS. Closing half the gap to proportional order share yields an estimated +BRL 1.5M in incremental annual revenue.

VIP CRM retention

Segment and re-engage at-risk customers. 10% re-conversion projects BRL 500K–750K in incremental annual revenue.

Instalment promotions

Reduce affordability friction in Health & Beauty, Watches & Gifts, and Bed & Bath to lift average order value.

Q4 surge planning

Pre-negotiate carrier capacity before Black Friday. November 2017 was the highest-revenue month and worst for on-time delivery.

Impact summary

BRL 2.25M+ estimated annual revenue uplift

+0.4 ↗ 44%

Faster delivery and proactive issue resolution drive higher customer ratings across the platform.

+15% ↗ 44%

Loyalty and re-engagement campaigns convert one-time buyers into returning customers.

+3 ↗ 44%

Seller diversification across new regions reduces single-state delivery risk and broadens reach.

All outputs live

Explore the work
open, documented, ready

1

Tableau Dashboard

Sales, delivery, and review trends. Live and publicly accessible.
Open Tableau Dashboard →

2

Live Web App

Full analytics platform built with React on Vercel and FastAPI on Render.
Visit the deployed app →

3

GitHub Repository

All notebooks, cleaned data, and model artefacts. Open and documented.
View on GitHub →

Everything is publicly accessible — open the links, run the models, fork the repo.